Lab 08: Inter-VLAN Routing

Router1 (2911) · Switch (2960) · PC0 · PC1 · PC2 · PC3

Lab #08	**Topic:** VLANs & Inter-VLAN Routing	**Tool:** Cisco Packet Tracer	**Cert:** CompTIA Network+

1. Lab Objectives

- Extend Lab 07 by adding Router1 (2911) to enable inter-VLAN routing.
- Configure a trunk link from the switch (Gig0/1) to Router1 (Gig0/0).
- Set up 802.1Q sub-interfaces on Router1 — one per VLAN.
- Connect PC0–PC3 to the switch and assign them to four VLANs.
- Verify full inter-VLAN communication via ping across all VLANs.

2. Network Topology

Router1 (2911) is on the left connected via Gig0/0 to the 2960-24TT switch on Gig0/1. The switch has PC0 (Fa0/1), PC1 (Fa0/2), PC2 (Fa0/3) and PC3 (Fa0/4) each on a separate VLAN. The router provides inter-VLAN routing via sub-interfaces.

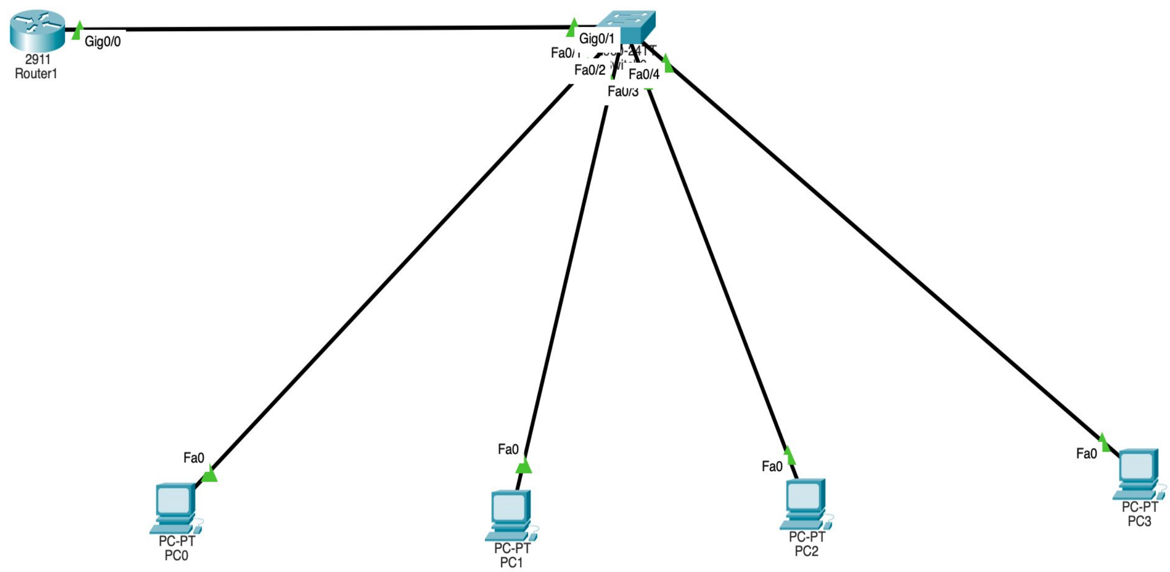

Figure 1 – Router1 (2911) connected to Switch0, with PC0–PC3 on separate VLANs

Device	Port	VLAN	IP Address	Gateway
PC0	Fa0/1 (Switch)	VLAN 10	192.168.10.10	192.168.10.1

PC1	Fa0/2 (Switch)	VLAN 20	192.168.20.10	192.168.20.1
PC2	Fa0/3 (Switch)	VLAN 30	192.168.30.10	192.168.30.1
PC3	Fa0/4 (Switch)	VLAN 40	192.168.40.10	192.168.40.1
Router1	Gig0/0 (trunk)	All VLANs	Sub-interfaces	—

3. Step-by-Step Configuration

3.1 Switch – VLANs and Trunk

```
Switch(config)# vlan 10
Switch(config-vlan)# name VLAN10
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name VLAN20
Switch(config-vlan)# exit
Switch(config)# vlan 30
Switch(config-vlan)# name VLAN30
Switch(config-vlan)# exit
Switch(config)# vlan 40
Switch(config-vlan)# name VLAN40
Switch(config-vlan)# exit
Switch(config)# interface Fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit
Switch(config)# interface Fa0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
Switch(config)# interface Fa0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
Switch(config-if)# exit
Switch(config)# interface Fa0/4
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 40
Switch(config-if)# exit
Switch(config)# interface GigabitEthernet0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
```

```
Switch# write memory
```

3.2 Router1 – Sub-interfaces

```
Router1(config)# interface GigabitEthernet0/0
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# interface GigabitEthernet0/0.10
Router1(config-subif)# encapsulation dot1Q 10
Router1(config-subif)# ip address 192.168.10.1 255.255.255.0
Router1(config-subif)# exit
Router1(config)# interface GigabitEthernet0/0.20
Router1(config-subif)# encapsulation dot1Q 20
Router1(config-subif)# ip address 192.168.20.1 255.255.255.0
Router1(config-subif)# exit
Router1(config)# interface GigabitEthernet0/0.30
Router1(config-subif)# encapsulation dot1Q 30
Router1(config-subif)# ip address 192.168.30.1 255.255.255.0
Router1(config-subif)# exit
Router1(config)# interface GigabitEthernet0/0.40
Router1(config-subif)# encapsulation dot1Q 40
Router1(config-subif)# ip address 192.168.40.1 255.255.255.0
Router1(config-subif)# exit
Router1# write memory
```

4. Verification

```
PC0> ping 192.168.20.10    ! VLAN10 to VLAN20
PC0> ping 192.168.30.10    ! VLAN10 to VLAN30
PC0> ping 192.168.40.10    ! VLAN10 to VLAN40
Router1# show ip route     ! 4 connected subnets via sub-interfaces
```

5. Key Takeaways

- Router-on-a-Stick uses ONE physical trunk link to route between all VLANs — cost-effective.
- The parent interface (Gig0/0) must be 'no shutdown' for all sub-interfaces to work.
- Each sub-interface handles one VLAN — encapsulation dot1Q tags the traffic correctly.
- Four VLANs means four sub-interfaces and four separate /24 subnets on the router.
- In production, a Layer 3 switch performs inter-VLAN routing in hardware — much faster.